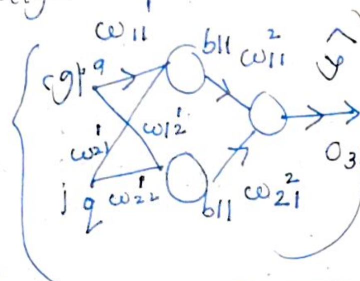
Backpropagation:

→ backprop of errors is an algo for sup. learning of AI. neural networks using gradient decent. Given an ANN & an neural network the method calculates the gradient of error function with respect to NN's weights.

(prerequisite: Grad. decent & forward prop

→ Matlab: its an algorithm to train Neural Network

cgpa	iq	lpa
8	80	8
7	70	7



b Linear

O_{12}

lay. No

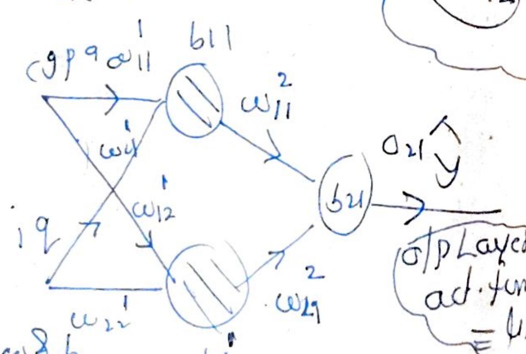
SN No

Backpropagation:

Hidden activation = Linear

w_{12}^1

cgpa	iq	lpa
80	8	3
60	9	5
70	5	8
120	7	11



o/p Layer act. func = linear

regression

steps ⓪ initialize w & b

① : you select a point / row / dp / student (1 row)

② predict (lpa) → f feed prop (dot product) ✱

① random values

② w = 1 b = 0 (we use it)

③ choose a loss function

(MSE lets say used here)

$(y_i - \hat{y}_i)^2$ ✱ lets say $\hat{y}$ = lpa

$MSE = (18-3)^2 = (15)^2 = \boxed{225}$ ✓

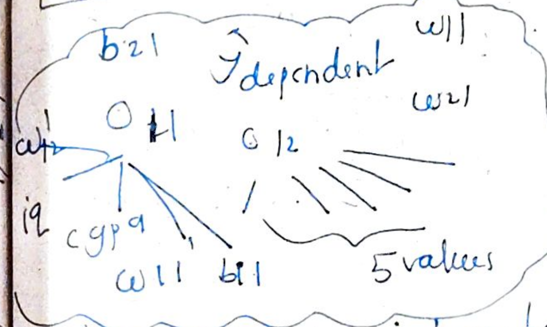
this loss function does: it shows our NN that what {dekh BKL tu itna jyada galat hai Lode}

$\boxed{O_{21} = \hat{y}_i = w_{11}^2 O_{11} + w_{21}^2 O_{12} + b_{21}}$

$\sigma(O_{21}) = \sigma(2) = O_{21}$ ✓

$\left\{\begin{matrix} w_{11}, & O_{12} \\ w_{21} & O_{12} \\ b_{21} & \end{matrix}\right\}$ → 5 parameters dependent

$\boxed{L_{min} \rightarrow \hat{y} \rightarrow O_{21}}$



→ this is why its called as back propagation of error ✱

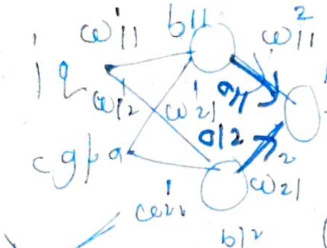
→ itne saari changes karne hai

④ weights & biases update (using GD)

$w_{new} = w_{old} - \eta \frac{\partial L}{\partial w_{old}}$	$b_{new} = b_{old} - \eta \frac{\partial L}{\partial b_{old}}$

apply to all trainable parameters ✓

→ Nikalo hathse and python me add karlo ✓



$L(min) \to \hat{y} \to w_{11}^2$

derivative $\left(\frac{dy}{dx}\right)$ (*)

(2)

(1)

$\boxed{\frac{\partial L}{\partial w_{11}^2}, \frac{\partial L}{\partial w_{21}^2}, \frac{\partial L}{\partial b_{21}}} \to \boxed{\frac{\partial L}{\partial w_{11}^1}, \frac{\partial L}{\partial w_{21}^1}, \frac{\partial L}{\partial b_{11}}}$

$\boxed{\frac{\partial L}{\partial w_{12}^1}, \frac{\partial L}{\partial w_{22}^1}, \frac{\partial L}{\partial b_{12}}} \to$ we need to calculate ✓

9 derivatives needed

$L = (y-\hat{y}_i)^2$ remember

$\to$ MSE used.

$$\frac{\partial L}{\partial w_{11}^2} = \left\{\frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial w_{11}^2}\right\}$$

$$\boxed{\frac{\partial L}{\partial \hat{y}} = \frac{\partial}{\partial \hat{y}}(y-\hat{y}_i)^2 = 2(y-\hat{y}_i)(-1)}$$

O_{21} (Hon)

$\hat{y} = O_{21} =$

$$\boxed{\frac{\partial \hat{y}}{\partial w_{11}^2} = \frac{\partial}{\partial w_{11}^2}(\underbrace{O_{11}w_{11}^2}+O_{12}w_{21}^2+b_{21}) = O_{11}}$$

$$\boxed{\frac{\partial L}{\partial w_{11}^2} = -2(y-\hat{y})O_{11}}$$ (1)

$\frac{\partial \hat{y}}{\partial b_{21}} = 1$

$$\boxed{\frac{\partial L}{\partial w_{21}^2} = -2(y-\hat{y})O_{12}}$$ (2) $\frac{\partial L}{\partial b_{21}} = \left\{\frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial b_{21}}\right\}$

$$\boxed{\frac{\partial L}{\partial b_{21}} = -2(y-\hat{y}) \cdot 1}$$ (3) $= -2(y-\hat{y})$

w_{11}^2 change aayega to

$\downarrow$

$\hat{y}$ me change aayega

$\downarrow$

L me change aayega

$\frac{\partial L}{\partial w_{11}^1}, \frac{\partial L}{\partial w_{21}^1}, \frac{\partial L}{\partial b_{11}}$

$\cancel{\frac{\partial L}{\partial w_{11}^1} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial w_{11}^1}}$

$\cancel{\frac{\partial L}{\partial w_{21}^1} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial w_{21}^1}}$

$\cancel{\frac{\partial L}{\partial b_{11}} = \left\{\frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial b_{11}}\right\}}$

$\cancel{\hat{y} \neq O_{11}}$

Here in step (2)

$\cancel{O_{11} \neq w_{11}^1}$

(b) (c)

(a) $\frac{\partial L}{\partial w_{11}^1}, \frac{\partial L}{\partial w_{21}^1}, \frac{\partial L}{\partial b_{11}} \to$ step (2)

(a)

$\boxed{w_{11}^2 O_{11} + w_{21}^2 O_{12} + b_{21}}$

$$\frac{\partial L}{\partial w_{11}^1} = \left\{\frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial O_{11}} \times \frac{\partial O_{11}}{\partial w_{11}^1}\right\}$$

$$\frac{\partial L}{\partial w_{21}^1} = \left\{\frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial O_{11}} \times \frac{\partial O_{11}}{\partial w_{21}^1}\right\}$$ (b) Nikalo

$\to$ V. imp

$$\frac{\partial L}{\partial b_{11}} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial O_{11}} \times \frac{\partial O_{11}}{\partial b_{11}}$$ $\to$ (c) ye hhi..

✓

lets calculate Now (*)

$\to$ common

Now for 3rd set of 3 derivatives.

$$\frac{\partial L}{\partial \hat{y}} = -2(y-\hat{y})$$

$$\frac{\partial L}{\partial \omega_{12}'} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial o_{12}} \times \frac{\partial o_{12}}{\partial \omega_{12}'} \quad d$$

$$\frac{\partial L}{\partial \omega_{22}'} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial o_{12}} \times \frac{\partial o_{12}}{\partial \omega_{22}'} \rightarrow e$$

$$\frac{\partial L}{\partial b_{12}} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial o_{12}} \times \frac{\partial o_{12}}{\partial b_{12}} \rightarrow f$$

→ similar

→ easy another

→ commar

→ 1 bar nikalo.

$X1 = cq$

$X2 = cgpa$

$$\boxed{o_{11} = \omega_{11}' X1 + \omega_{21}' X2 + b_{11}}$$

$$\boxed{\hat{y} = \omega_{11}^2 o_{11} + \omega_{21}^2 o_{12} + b_{21}}$$

→ yaad karo

$$\boxed{\frac{\partial \hat{y}}{\partial o_{11}} = \omega_{11}^2} = \boxed{\frac{\partial \hat{y}}{\partial o_{12}} = \omega_{21}^2}$$

easy hai diffrenhiation

$$\boxed{\frac{\partial o_{11}}{\partial \omega_{11}'} = X1} \quad \boxed{\frac{\partial o_{11}}{\partial \omega_{21}'} = X2} \quad \boxed{\frac{\partial o_{11}}{\partial b_{11}} = 1}$$

$$\boxed{\frac{\partial o_{12}}{\partial \omega_{12}'} = X1} \quad \boxed{\frac{\partial o_{12}}{\partial \omega_{22}'} = X2} \quad \boxed{\frac{\partial o_{12}}{\partial b_{22}} = 1}$$

$$\boxed{o_{12} = \omega_{12}' X1 + \omega_{22}' X2 + b_{22}}$$

$$d = \frac{\partial L}{\partial \omega_{12}'} = -2(y-\hat{y}) \times \omega_{21}^2 \times X1 \quad (2)$$

$$e = \frac{\partial L}{\partial \omega_{22}'} = -2(y-\hat{y}) \times \omega_{21}^2 \times X2 \quad (8)$$

$$f = \frac{\partial L}{\partial b_{12}} = -2(y-\hat{y}) \times \omega_{21}^2 \times 1 \rightarrow (9)$$

$$a = \frac{\partial L}{\partial \omega_{11}'} = -2(y-\hat{y}) \times \omega_{11} \times X1 \rightarrow (4)$$

$$b = \frac{\partial L}{\partial \omega_{21}'} = -2(y-\hat{y}) \times \omega_{11}^2 \times X2 \rightarrow (5)$$

$$c = \frac{\partial L}{b_{11}} = -2(y-\hat{y}) \times \omega_{11}^2 \times 1 \rightarrow (6)$$

we have calculated equations (1) (2) (3) (✓)

(9) derivatives (✓)

$$\boxed{\frac{\partial L}{\partial \omega_{11}^2} = -2(y-\hat{y}) o_{11}} \; (1)$$

$$\boxed{\frac{\partial L}{\partial b_{21}} = -2(y-\hat{y}) \cdot 1} \; (3)$$

$$\boxed{\frac{\partial L}{\partial \omega_{21}^2} = -2(y-\hat{y}) o_{12}} \; (2)$$

I am happy